

OLIVIA, MN, USA

WMO: 722168

Lat: 44.779N

Lon: 95.033W

Elev: 328

StdP: 97.45

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04960

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.8	-22.2	-27.5	0.3	-23.4	-25.9	0.4	-22.0	14.4	-11.5	12.7	-9.9	3.3	320	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	32.2	23.0	29.9	21.5	27.7	20.2	24.3	29.3	23.0	27.6	22.0	26.1	5.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	17.9	27.7	22.0	17.3	26.7	20.9	16.2	25.5	74.8	29.6	69.7	27.6	66.1	26.5	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.6	9.1	-28.5	34.8	2.5	1.5	-30.3	35.8	-31.8	36.7	-33.2	37.6	-35.1	38.7		
(5)			-28.8	26.7	2.5	1.2	-30.5	27.6	-32.0	28.3	-33.4	29.0	-35.2	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-9.1	-8.3	-0.2	7.1	13.9	19.6	21.7	19.7	16.5	8.5	1.3	-6.9	(6)
(7)		DBStd	12.26	7.13	6.82	6.89	5.71	5.06	3.39	3.12	3.10	4.54	5.21	5.98	6.54	(7)
(8)		HDD10.0	2452	593	513	322	119	19	0	0	0	4	91	267	524	(8)
(9)		HDD18.3	4402	852	746	574	338	153	24	6	20	88	307	512	782	(9)
(10)		CDD10.0	1377	0	0	6	31	142	287	363	301	198	46	4	0	(10)
(11)		CDD18.3	287	0	0	1	1	17	61	111	63	32	3	0	0	(11)
(12)		CDH23.3	2475	0	0	2	13	207	526	948	466	275	38	1	0	(12)
(13)		CDH26.7	653	0	0	0	2	64	146	268	99	68	7	0	0	(13)
(14)	Wind	WSAvg	4.1	4.6	4.5	4.5	4.9	4.6	3.8	3.0	2.8	3.6	4.2	4.4	4.4	(14)
(15)	Precipitation	PrecAvg	621	17	15	34	62	76	98	85	83	67	49	32	15	(15)
(16)		PrecMax	895	118	53	136	158	184	305	215	199	176	152	94	44	(16)
(17)		PrecMin	365	0	0	6	8	11	5	15	17	13	1	0	1	(17)
(18)		PrecStd	127	19	12	24	33	37	55	45	37	39	37	28	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	12.0	21.1	25.1	32.4	32.5	33.8	32.3	32.3	27.4	20.0	10.1	(19)
(20)			MCWB	2.8	9.5	14.8	15.0	19.6	22.2	26.1	24.6	21.9	16.9	13.0	7.8	(20)
(21)		2%	DB	2.6	5.9	16.1	22.3	27.7	31.0	32.1	28.9	27.9	23.0	15.2	5.1	(21)
(22)			MCWB	1.4	3.3	11.3	13.7	17.2	21.4	24.4	22.6	19.8	14.6	10.1	2.3	(22)
(23)		5%	DB	1.3	2.5	12.3	19.9	25.2	27.8	29.9	27.4	26.5	20.9	12.3	2.4	(23)
(24)			MCWB	0.6	1.2	9.6	12.1	16.1	19.4	22.7	20.9	18.7	14.1	8.0	1.3	(24)
(25)		10%	DB	0.2	1.1	9.8	16.2	22.7	26.3	27.6	26.1	24.8	17.0	10.0	1.2	(25)
(26)			MCWB	-0.1	0.3	7.3	10.1	15.5	18.3	21.0	20.0	18.4	11.5	6.8	0.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.3	9.1	15.8	16.2	21.4	24.8	27.2	25.4	23.5	20.3	13.9	8.0	(27)
(28)			MCDWB	5.7	11.7	20.1	23.7	28.5	30.9	31.9	31.1	29.1	23.8	18.4	9.2	(28)
(29)		2%	WB	1.5	3.4	11.9	14.6	19.4	22.5	25.1	23.6	22.0	16.8	10.5	3.0	(29)
(30)			MCDWB	2.6	5.6	14.2	20.5	24.9	27.6	30.5	27.9	26.4	22.5	14.6	4.5	(30)
(31)		5%	WB	0.7	1.3	9.2	12.8	17.9	21.3	23.5	22.3	20.6	14.1	8.4	1.4	(31)
(32)			MCDWB	1.2	2.3	12.5	18.1	23.6	26.3	28.1	25.9	24.2	18.7	11.6	2.3	(32)
(33)		10%	WB	-0.2	0.2	6.6	11.1	16.0	19.8	22.3	21.3	19.2	12.2	6.6	0.5	(33)
(34)			MCDWB	0.1	0.9	9.5	15.6	21.6	24.7	26.6	24.4	23.4	16.0	9.9	1.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.5	9.3	8.6	10.6	11.5	10.4	10.9	10.7	11.8	10.6	9.3	7.9	(35)
(36)			MCDBR	8.9	10.4	13.9	15.8	15.1	12.6	12.0	12.1	13.8	15.7	13.8	9.4	(36)
(37)			MCWBR	7.9	8.6	9.3	9.4	7.7	6.2	6.5	6.7	7.1	8.8	8.8	7.9	(37)
(38)		5% WB	MCDBR	8.5	9.7	13.2	13.7	12.4	10.7	10.8	10.3	11.4	12.7	12.8	8.4	(38)
(39)			MCWBR	7.7	8.2	9.2	8.8	7.3	6.4	6.4	6.7	7.1	8.0	8.7	7.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.258	0.277	0.306	0.363	0.384	0.395	0.401	0.408	0.366	0.318	0.280	0.259	(40)	
(41)		taud	2.468	2.414	2.448	2.338	2.338	2.362	2.334	2.294	2.387	2.520	2.542	2.476	(41)	
(42)		Ebn at Noon	881	920	934	898	885	874	864	843	854	861	848	843	(42)	
(43)		Edh at Noon	68	87	97	118	122	120	122	123	103	78	64	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.81	2.86	3.79	4.69	5.41	6.03	6.50	5.38	4.19	2.72	1.76	1.32	(44)	
(45)		RadStd	0.21	0.33	0.37	0.45	0.41	0.44	0.23	0.43	0.31	0.37	0.16	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (66 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page